BDSD -review

---

1. ch01 big data intro

1.1 background

1. revolution

1st steam era : promote the development of machinery industry

2nd electric era :automation level

3rd information era : transformation and upgrading of enterprises

4th big data era : intelligent processing of enterprises

2. information revolution

IT era -> Internet era -> Mobile Internet era-> Big data era-> AI era ->IoT era (物联网)

4 waves : big data, AI ,IoT,5G

3 elements : internet, cloud computing , big data

1.2 value

1.3 concepts

1. features

volume : huge amount of data

velocity: change fast

variety : various type , unstructured and structured data.

veracity: low value density

2. process flow

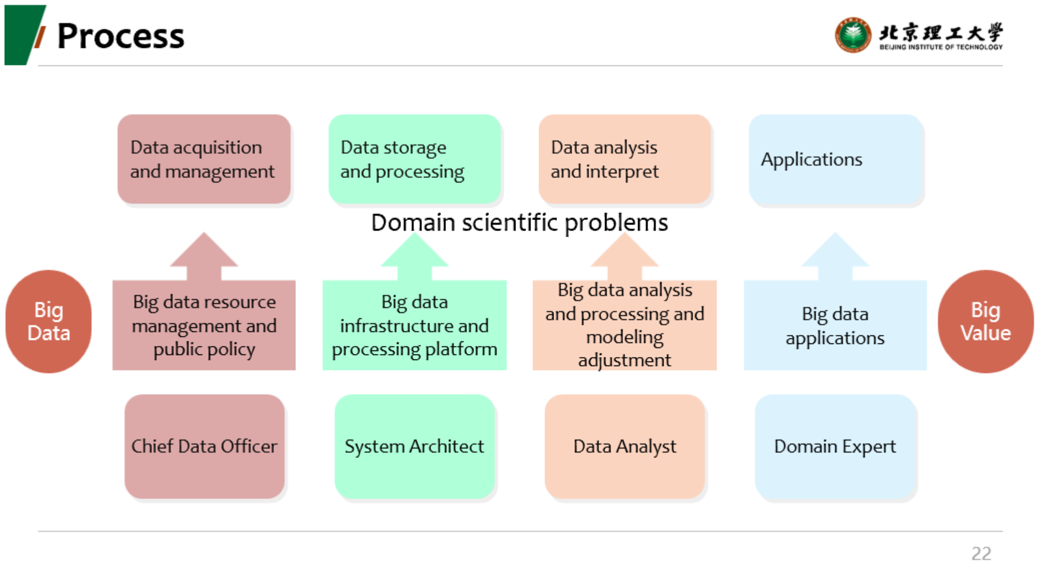


Figure 1

3. interdisciplinary 跨学科的

computer science, math and statistics, domain knowledge, ml , etc

4. science paradigms 科学研究的四个范式

experimental thinking, 实验 经验主义 empirical describing the phenomena

logical thinking , theoretical using models and generalizations

computational thinking, simulating complex phenomena

data thinking, data-intensive, unify theory, experiment, and simulation.

2. ch02 system intro

2.1 software framework

2.1.1 Hadoop

open-source software for reliable, scalable distributed computing.

Apache Hadoop software library is a framework, distributed processing of large data sets across clusters of computers using simple programing models.

scale up from single server to thousands of machine(each provide storge and computing), not rely on the hardware, it is designed to detect error in application layer, delivering a highly- available services in top of cluster (but each may prone to error).

JAVA, HDFS, MapReduce.

history : become the fastest system in 2008, for sorting the TB data.

ecosystem 生态

1. HDFS: Hadoop distributed file system: primary storage. storage layer of Hadoop.
2. MapReduce: the data processing system.
3. YARN: source management.

※ HDFS: written in JAVA . the primary storage.

provide

1. fault tolerance: hardware failure.
2. cost-efficient: simple coherency model, moving computation is cheaper than moving data.
3. large scale: streaming data access and large data sets.

   also, portability(可转移性) to the heterogeneous (异质的) soft and hard ware.

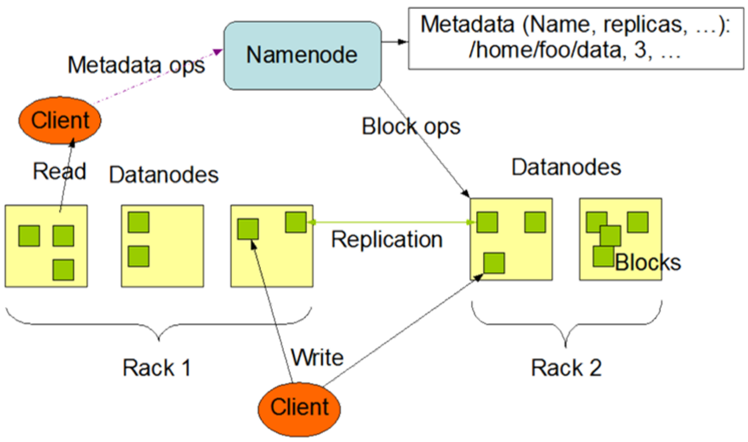


Figure 2

HDFS use the master-slave model

1. master is the Namenode, usually high-end machine , store the metadata , manage the file system namespace, regulate client's access and file system execution(naming, closing , opening file, directories)
2. slave is the Datanode, usually inexpensive computer, store the actual data, perform operations like block replica (复制品) creation, deletion and replication according to the instruction of Name Node. It manages data storage of the system.

※ MapReduce

process vast amounts of data in-parallel(并行) on large clusters of commodity hardware in a reliable , fault-tolerant manner.

working:

1. map phase: input data -> key-value pairs.
2. reduce phase: aggregation based the map phase output, such as 词频统计

example

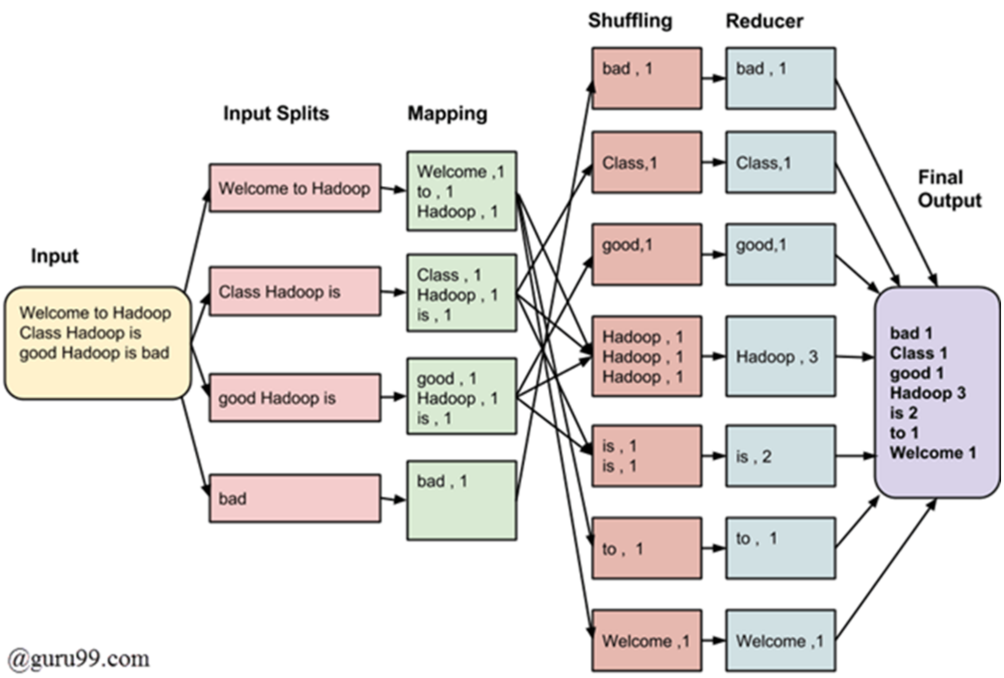


Figure 3

input splits: 切分

mapping: counting the occurrence

shuffling: club the same words

reducing: reducer aggregate the output value of shuffling phase

※ YARN yet another resource negotiator (谈判家)

provide resource management, managing and monitoring workloads .

working:

split up the functionalities(功能) of source management and job scheduling/ monitoring into separate daemons(进程).

global resource manager RM

per- application Application-Master AM

application: single job or DAG of job.

components:

1. resource manager : on master , how much resources each slaves have , scheduling the assignment to various tasks and keep track of node manager's heart beat.
2. node manager : on slaver, manage containers, utilization the resources and send heart beat to resource manager. (container: a fraction of node resource capacity)

features:

fault-tolerance, high availability, high processing, data logicality, data reliability.

2.1.2 Spark

unified analytics engine for large-scale data processing.

provide high-level APIs in java, Scala,python, and R. optimized engine support general execution graph.

spark SQL , MLlib, GraphX…

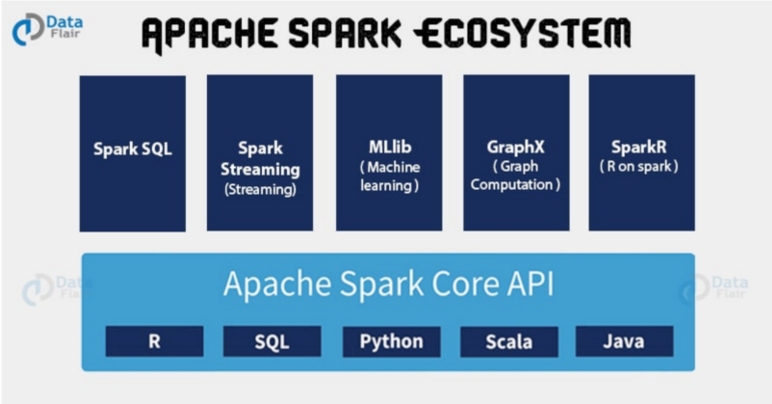


Figure 4

1. spark core.

foundation of parallel and distributed processing of huge dataset.

all function provided by spark are built on the top of spark core.

RDD: resilient(弹性的 可复原的) distributed dataset

1. fault tolerance, data can be operated in parallel
2. partitioning data across all node in cluster.

※ application

1. spark SQL

distributed framework for SQL

enable users to run SQL/HQL queries , provide Spark with more information about data and computation.

enable powerful, interactive , analytical application across both streaming and historical data.

involve dataframe and dataset to overcome the limitations with RDD

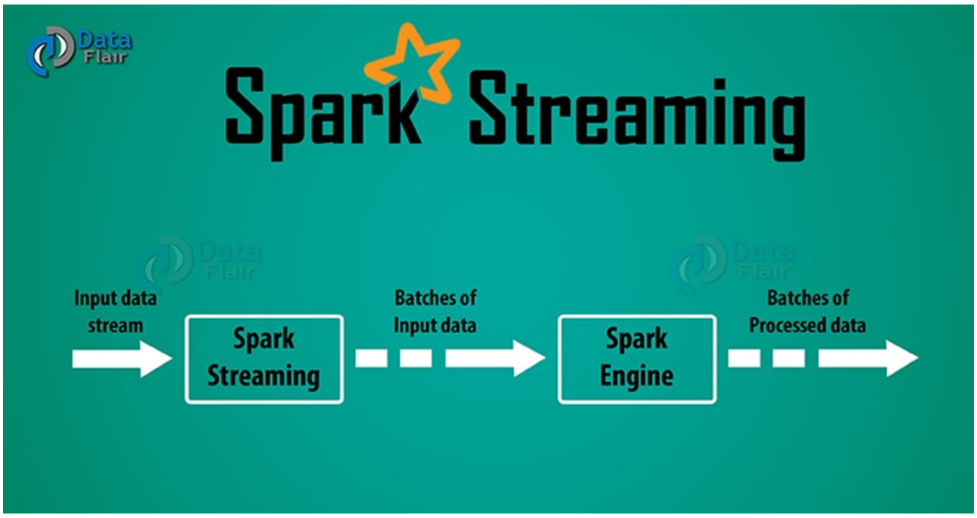
2. Spark Streaming

an add-on to core Spark API which allows scalable , high-throughput, fault-tolerance stream processing of live data streams.

micro-batching techniques, allow a process or task to treat stream as a sequence of small batch of data.

Spark turn stream into sequence of small batches of data.

过程:



gathering -> process-> storage.

A discretized stream or D-stream represent the stream of data divided into small batches, built on Spark RDD

3. SpaRk MLlib

scalable machine learning library, allows high-quality algorithm and speed.

contains various algorithm, clustering, regression , classification and topic modeling(LDA)

dataframe-based API is the primary machine learning API.

4. GraphX

graph-parallel computation

network of graph analytics and data store

algorithm : PageRank Connected components Label propagation SVD++

features: 1. flexibility, seamlessly work with both graph and collections. 2. speed.

※ Spark features

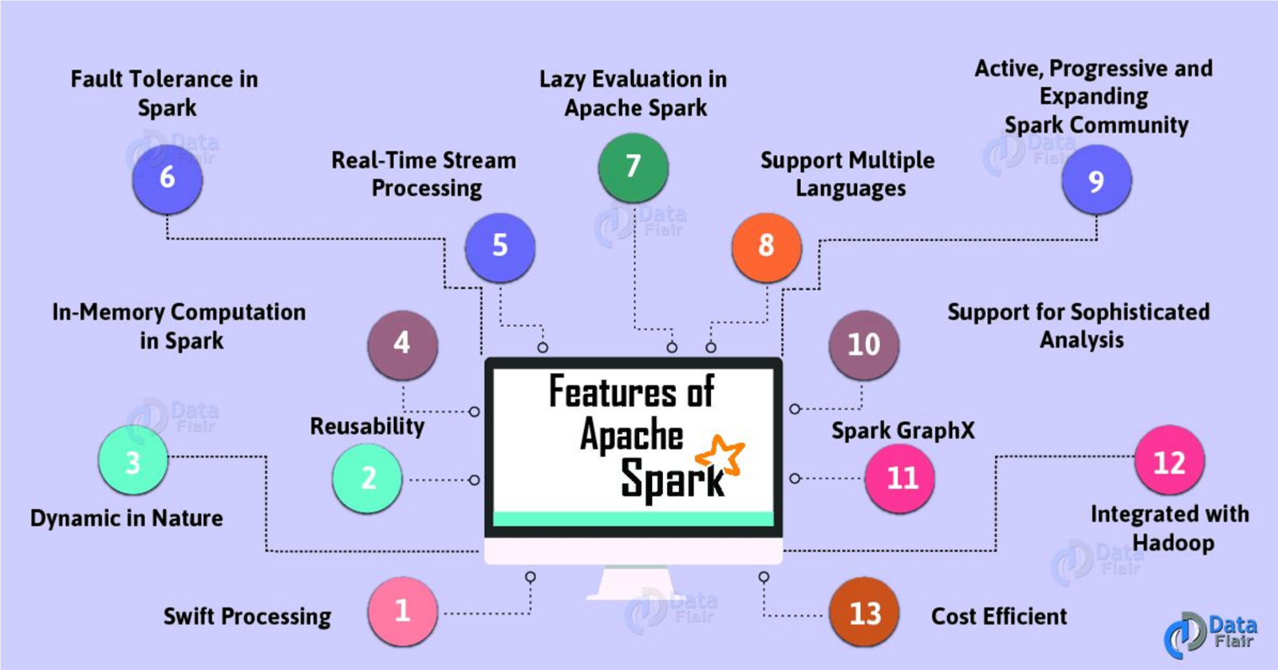


Figure 5

2.1.3 Storm

free and open-source distributed real-time computation system.

easy to reliably process unbounded (infinite) stream of data, doing for real-time processing what hadoop do with batch of data.

1. components

tuples : core unit of data, immutable set of key-value pairs

stream: unbounded stream of tuples.

spouts: sources of stream, wrap a streaming data source and emit tuples.

bolts: core functions of streaming computation, receive tuples and do stuff(read ,write to a data store and do arbitrary computation 任意的)

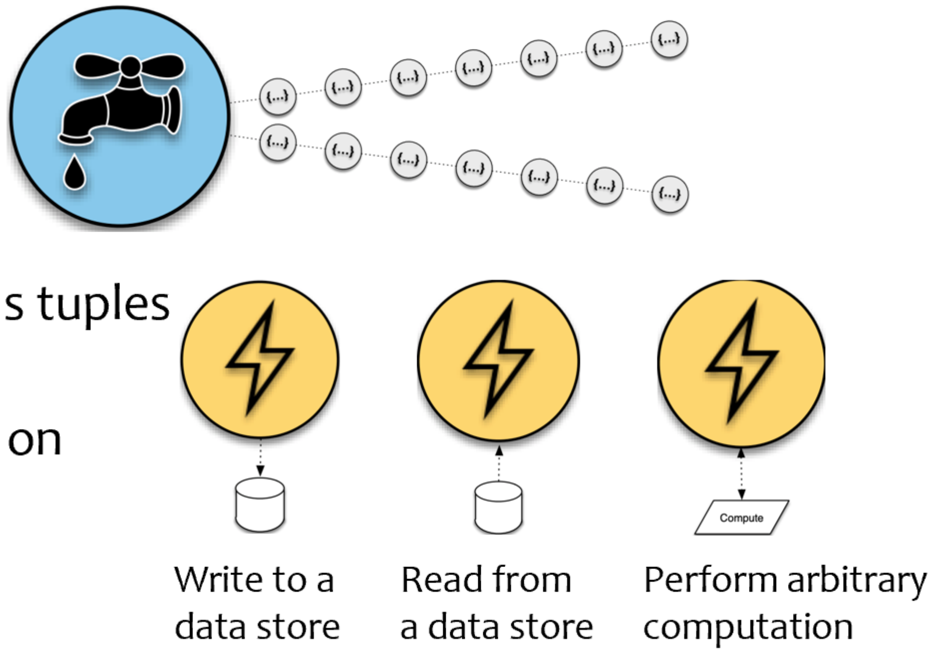


Figure 6

2. topology 拓扑结构

DAG 有向图 data flow representation and streaming computation.

storms execute spouts and bolts as individual tasks in parallel on multiple machines.

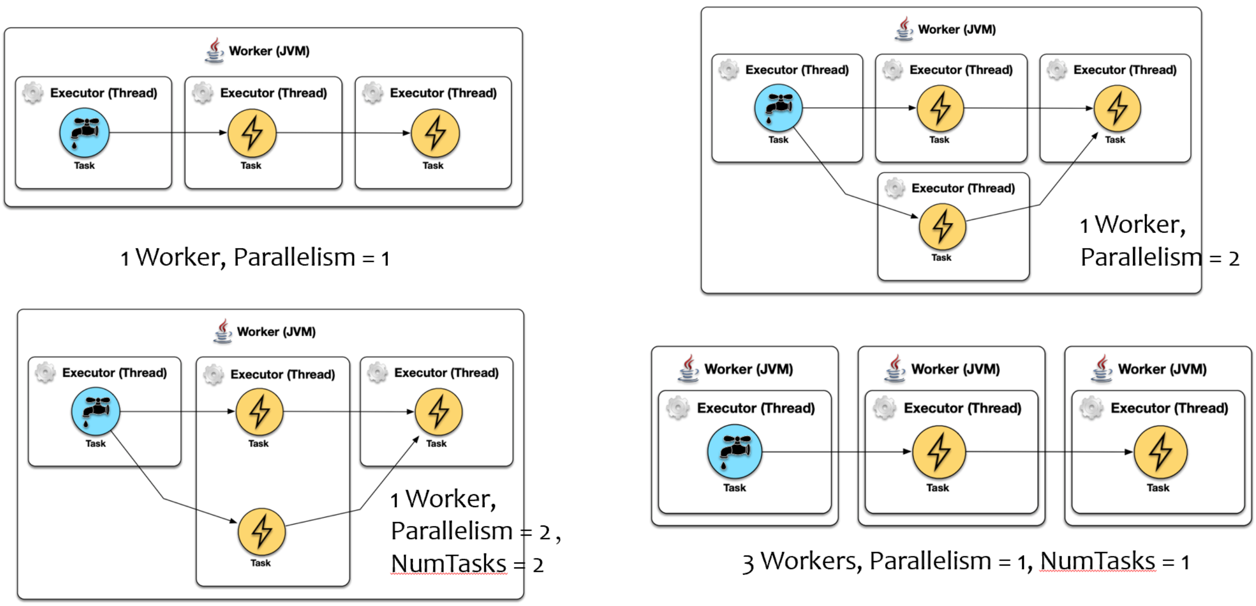


Figure 7

3. features

real-streaming: key enabler of lambda architecture

fast:

scalable : cluster

fault-tolerance:

reliable: guarantee the message delivery and exactly-once semantics (语义学).

2.2 development workflow

1. requirement : data input and output
2. data volume ,process efficiency and reliability
3. data modeling
4. architecture design
5. consider the interaction between big data system and enterprise IT system.
6. Finalize selection and specification.
7. write basic service code based on data modeling
8. formally write the first module
9. realize other modules and compute testing and debugging.
10. testing and acceptance.

3. ch03 Data acquisition and integration

3.1 concepts

1. data

data: a units of information, often numeric , that are collected through observation.

data is qualitative and quantitative description of things.

data are set of values of qualitative or quantitative(定性或定量) variables about one or more persons or objects.

datum (singular of data) is single value of a single variable.

type : numeric (continuous), categorical(discrete), text, image, XML.

the generation of big data is the result of development of computer and internet.

IoT data :物联网 a smart city is an urban area that use different type of machine to supply information to manage assets and resources efficiently.

2. challenges 大数据面临的挑战

various data type , including unstructured data.

large volume of data

reliability and efficiency of acquisition/ collection.

how to avoid data duplication

how to ensure data quality.

3. data-intensive society

3.2 data source

3.2.1 sources

Human with computer

1. management information service MIS

internal information system in cooperation, structured data.

2. web information system

internal/ external information system in the internet, social network, social media , semi-structured / unstructured data.

Object with computer

3. physical information system PIS

on various physical objects and physical process, such as real-time monitoring , detection,

used for production scheduling, process control , environmental protection.

measurement value of physical, chemical, biological.

multimedia data, audio, video

4. scientific experiment system

belong to the PIS, but the experimental environment is pre-set and mainly for research.

data are selected and under control.

synthetic data by simulation. 合成的/ 人造的

3.2.2 other categories

1. ways of collection

offline: collect data at a certain intervals 间隔区间 but not too short

online : collect data once the data generates or arrives.

2. where the data from

internal data: platform of own and historical data. easier to get .

external data: data in web

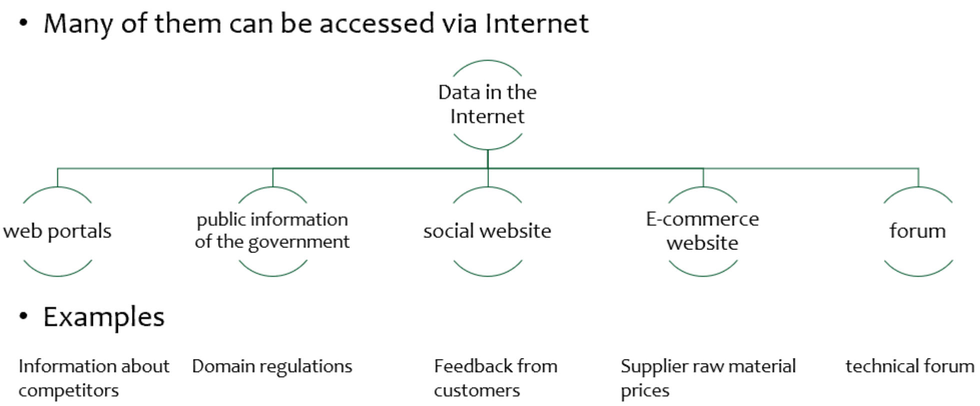


Figure 8

3.3 more introduction

3.3.1 offline data internal data

1. internal ETL

E: extract , read data from multiple data resource, and extracts required set of data.

recover necessary data with optimum usage of resources.

should not disturb (打扰) data sources, performance, and functioning

T: transform, filtration(过滤), cleansing , and preparation of data extracted, with lookup tables.

authentication of record (认证), refutation (反驳), and integration of data.

data to be sorted, filtered, cleared, standardized, translated or verified for consistency.(一致性)

L: load, writing data output,after extraction and transformation to a data warehouse(仓库).

physical insertion of record as a new row in database,

link processes for each record from the main source.

2. data extraction

※ full extraction : extract all the data in the data source. database.

convert them into the format that can be recognized by ETL tools.

will include duplicated data since history data will be extracted many times.

※ incremental extraction

only extract the data updated/ newly added in db from last extraction.

should not cause too much pressure on existing business system.

method: log comparison, timestamp, trigger, and table difference.

1. log comparison: based on the log of db to find the changed data.

tool: Oracle CDC(changed data capture)

CDC can identify the data that has changed since the last extraction

use CDC while extracting ,updating or deleting the source table,

the changed data is stored in the changed table of DB

two types:

synchronous : using triggers that inserts entries to a Change Capture table every time a data modification operation is executed.

asychronous : using Redo Logs that keep a track of all activities in a database.

2. timestamp

add a timestamp field(attribute) and modify the value of the corresponding timestamp , while changing the data.

when extracting data, determine which data to extract by comparing the value of the system time and the timestamp time.

for DBs that support automatic update of timestamp, when other fields in DB table change , the system automatically updates the value of the timestamp.

pros: has a good performance, easier to extract data.

cons: inject(注射)extra data to the business system ,add the timestamp

the deletion and update operations of data before the timestamp cannot be captured, and the data accuracy is subject to certain restriction.

3. triggers

create trigger for insert and update

once the corresponding data in the table changes , the changed data will be written into temporary table by trigger.

The extraction thread extracts data from the temporary table.

The extracted data is marked or deleted.

pros: better data extraction performance

cons: have an impact on the business system.

4. table differencing

using MD5 check code, create an MD5 temporary table with a similar structure for the table to be extracted.

what the temporary table record : the primary key of the source table and the MD5 check code calculated based on the data of all fields .

each time data is extracted, compare source table and the MD5 temporary table with the MD5 check code to determine whether the data in the original table is inserted, updated or deleted. update the MD5code at same time.

cons: when there is no primary key or unique column in the table, and contains duplicate records , the accuracy of the MD5 is poor.

※internal transform and load

从数据源中抽取的数据不一定会完全满足目的数据库的要求，例如格式，数据输入错误，数据不完整。

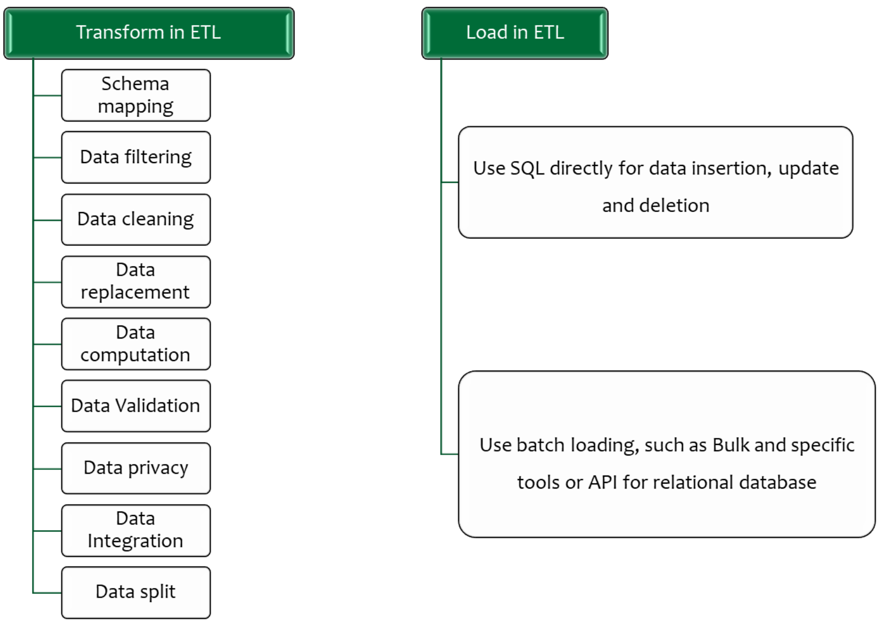


Figure 9

3.3.2 offline data external data

1. Network big data

big data generated by the interaction and integration of the ternary world (三元世界) of human, machines and things in cyberspace (赛博空间) and available on the internet .

features: high noise , burstiness (突发性), interactivity , sociality , timeliness , heterogeneous (异质的).

2. web crawler

spider , an internet bot that systematically browses the world wide web, typically operated by search engine for purpose of web indexing.

High level architecture of a spider: seed, crawl frontier , policies, repository

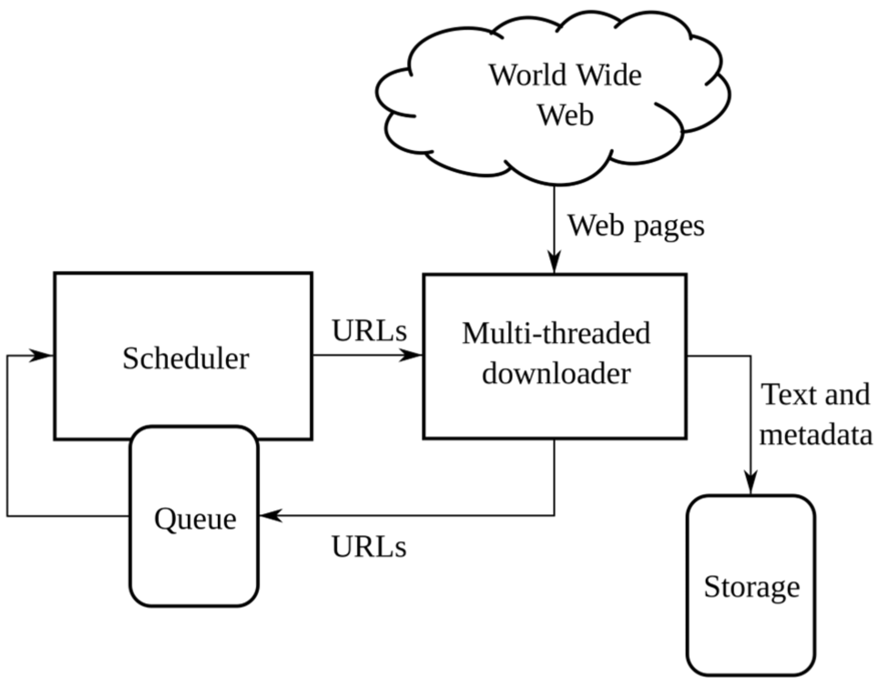


Figure 10

seeds : the list of URLs

crawl frontier : visit all the hyperlink in the page an add in the list of URLs called crawl frontier.

URLs from the seeds are called recursively , according to the policy.

If the crawler is performing archiving (网页归档), it copies and save the page as it goes, in such way they can be viewed , snapshots (快照)

repository: archive, is designed to store and manage the collection of web pages.

The repository only store the HTML file in distinct file, is similar to the modern-day DB. but difference, does not need all the functionally offered by a database system, store the most recent version of the page.

crawling policy:

selection policy, which states the pages to download

re-visit policy which states when to check for changes to the page

a politeness policy , how to avoid overloading web site

a parallelization policy, how to coordinate distributed web crawlers

※ type

1.focused web crawler: selectively search for web pages , to obtain more relevant information with high level of accuracy

2.incremental web crawler ,designed to visit and access updated web page

3.distributed web crawler, assign crawling to other crawler, A central server in remote areas communicates and syncs with the nodes.

4.parallel web crawler, multiple mechanisms of the parallel crawler are created and used that operate parallel

5.hidden web crawler: these pages are referred to as the hidden web or the deep web.

※ crawl policy DFS and BFS

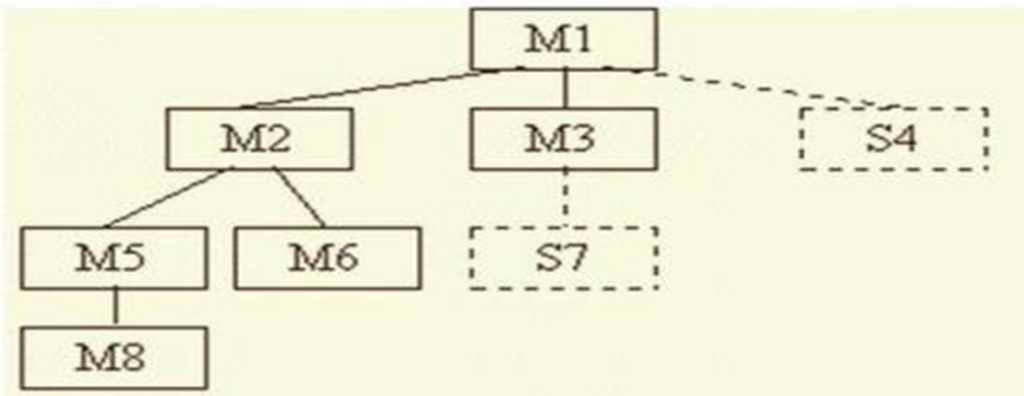


Figure 11

policy- page rank

an algorithm used by Google to rank web page in their search engine result.

works by counting the number and quality of links to page to determine a rough estimate.

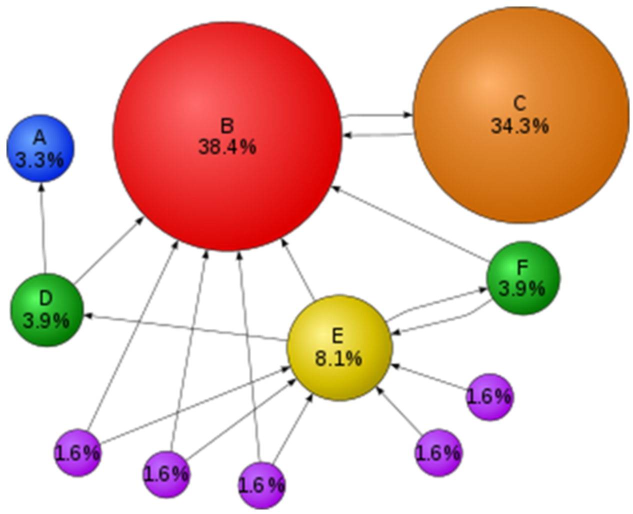


Figure 12

3.3.3 offline data deep web

The deep web, invisible or hidden web are parts of www , whose contents are not indexed by standard web search engine.

1. contents

pages that are not referred to by search engines due to lack of directed links

Non-HTML file accessible on the web, such as image, pdf and word document.

A dynamic page obtained by querying the back-end online database by filling in the form

content that requires registration or other restrictions to access.

2. difference with deep web and search engine

API: complex API for web database need query in deep wed but search engine only need the key word search

result: structured data for deep web, but web page for search engine search result order, certain attribute value in deep web for deep web, similarity between search result and query for search engine.

3.

multiple ways to parse html form into database

associate html forms with specific fields to realize automatic filling of forms.

domain-independent detection 领域无关检测

不依赖领域知识，而是通过通用策略来进行检测，以较小的查询次数获得更多的结果 iterative

3.4 online data

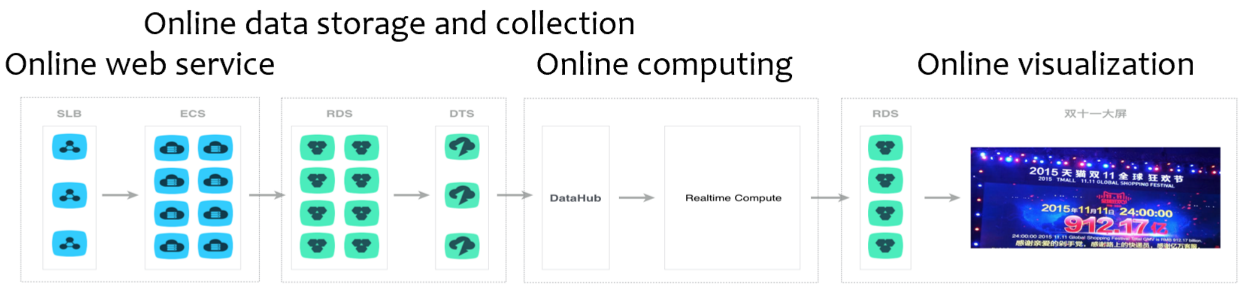


Figure 13

tools Flume

distributed reliable and available service for efficiently collecting, aggregating and moving large amount of log data.

simple and flexible architecture based on streaming data flows

robust and fault tolerant , with tunable reliability mechanisms and many failover and recovery mechanisms.

uses simple extensible data model that allows for online analytic application.

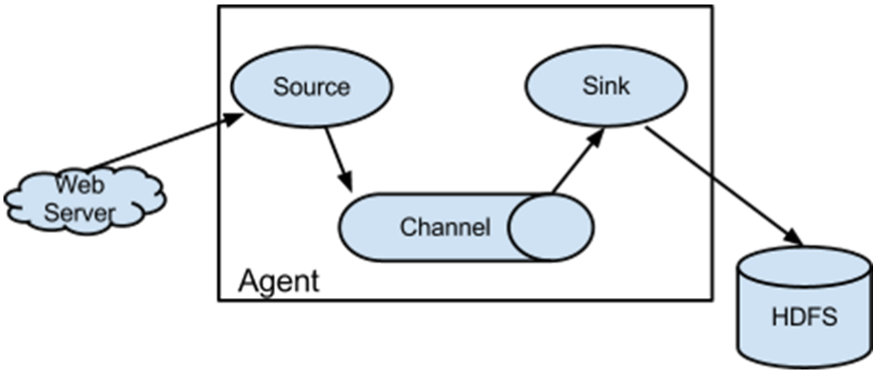


Figure 14

architecture

1. event : unit of data that flows through a Flume agent, from Source to Channel to sink
2. Agent: a process JVM that hosts the components that allow events to flow from an external to an external destination.
3. source

consume events having a specific format, and those events are delivered to the source by an external source like a web server.

when a source receives an event, it stores it into one/ more channel.

4. channel: a passive store that holds the event until that event is consumed by sink
5. sink: responsible for removing an event from the channel an putting it into an external repository or forwarding it to the source at next hop of the flow.

the source and sink within given agent run asynchronously with the event staged in the channel.

here is the pros and cons:

1. pros

enable us to store streaming data into any of the centralized repositories(HBase HDFS)

provide steady data flow between producer and consumer during reading and writing operations.

supports the feature of contextual routing

guarantees reliable message delivery

2. cons

weaker ordering guarantees

does not guarantee the message reaching 100% unique (duplicated)

complex topology and reconfiguration is challenging (配置)

suffer from scalability and reliability issue (规模和可靠性)

application：

e-commerce company, move huge amount of data , fraud detection (欺诈), loT applications, aggregating machine and sensor-generated data, alerting or SIEM

SIEM: security information and event management.

4. ch04 storage and management

storage and management are key procedure of big data processing

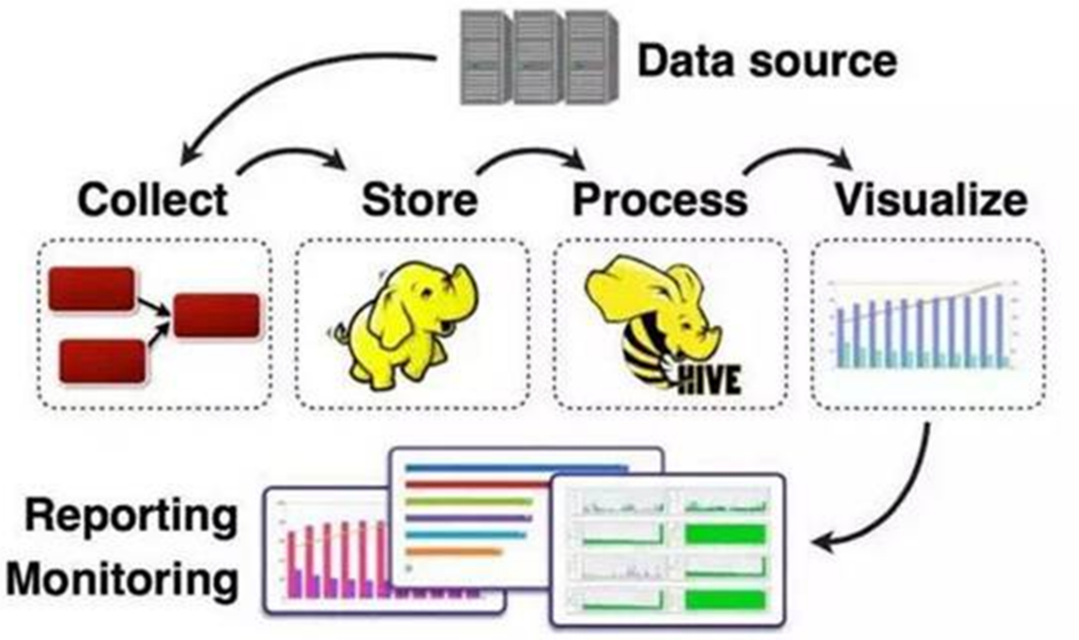


Figure 15

the original relational model have difficulty to deal with big data.

unstructured

数据模式不够灵活

4.1 distributed file system

DFS is distributed on multiple file servers or multiple locations, allows programs to access or store isolated files as they do with the local ones, allowing programmers to access files from any network or computer. 分布存储在多个计算机节点上

1. pros

low cost, easy to expand, robust and reliable, high availability.

user don't need to know where the data is , they can store and manage data like using local file system.

2. judgement how to evaluate

data storage method : the distribution strategy of files among various nodes

reading rate,

data security mechanism , adopt redundancy , backup mirroring and other methods to ensure the data security.

负载均衡， 备份数据一致性。

在一个节点存储m文件，在其他n个节点各村m/n作为备份。

2. major DFS

GFS(google file system): Linux-based proprietary(所有的), distributed system

HDFS : Hadoop

Geph : open-source distributed storage project , developed rapid in china

Lustre : large-scale , sage and reliable cluster file system, developed by SUN

TFS : Taobao File System, for massive unstructured data.

3. HDFS

HDFS provide storage for massive amounts of data.

MapReduce provide calculations for massive amounts of data.

※ features :

1. compatible with cheap hardware, tolerant to the hardware failure
2. streaming data access, instead of random read and write, can meet the requirement of large-volume data and improve throughout.
3. storage and management of large files
4. simple file model , write once read many times.
5. strong cross-platform compatibility

可靠性，安全性，高可用性

一个文件经过创建，写入和关闭之后就不要改变，不适合低延迟数据访问，无法高效存储大量的小文件，不支持文件随机修改，仅支持追加写入。

※ architecture

master/slaver architecture.

a single NameNode : master

a number of DataNode : slaver, usually one per node in the cluster, manage storage.

a file is split into one or more blocks and these blocks are stored in a set of Datanode

Clients can access these data blocks in parallel, greatly improve the access speed.

※data replication 复制

HDFS store each file as a sequence of blocks , the blocks of file are replicated for fault tolerance. the block size and replication factor are configured per file .

For common case, when replication factor = 3

if the writer is on datanode, 1 for local machine

else one on random datanode in the same rack

2nd for remote rack and 3rd for same remote rack

cuts the inter-rack write traffic(2,3 for same rack)

reliability and availability guarantee.

but not reduce the aggregate network bandwidth and replicas are not balanced

※ Read data

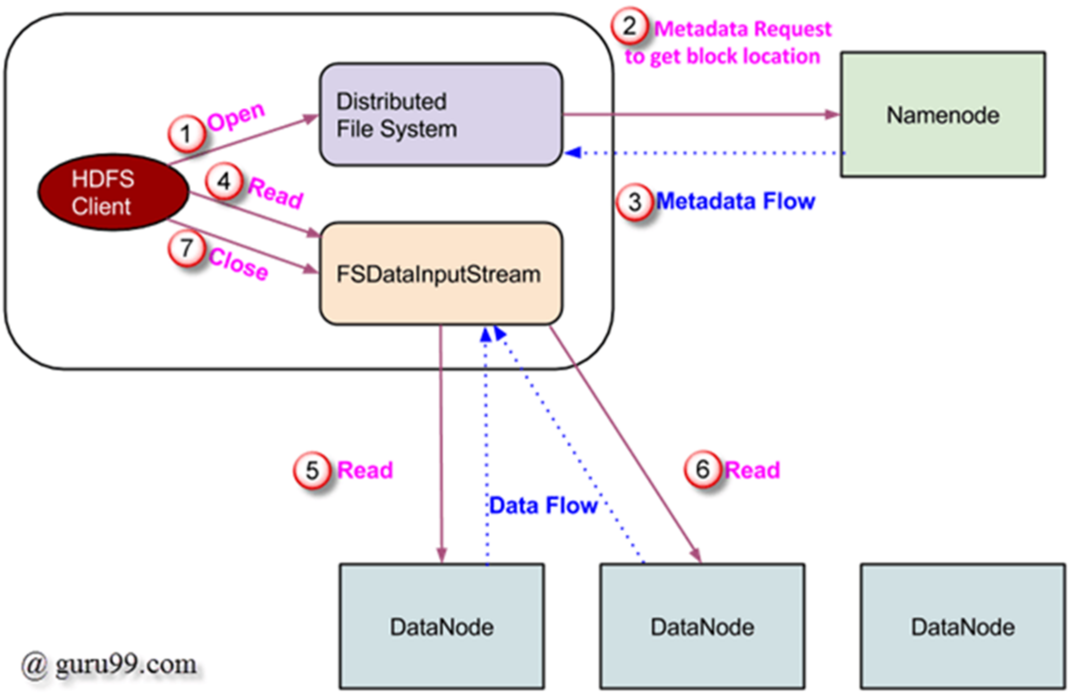


Figure 16

open-> metadata request-> return address-> read -> read repeatedly -> move to locate the next -> close

※ Write data

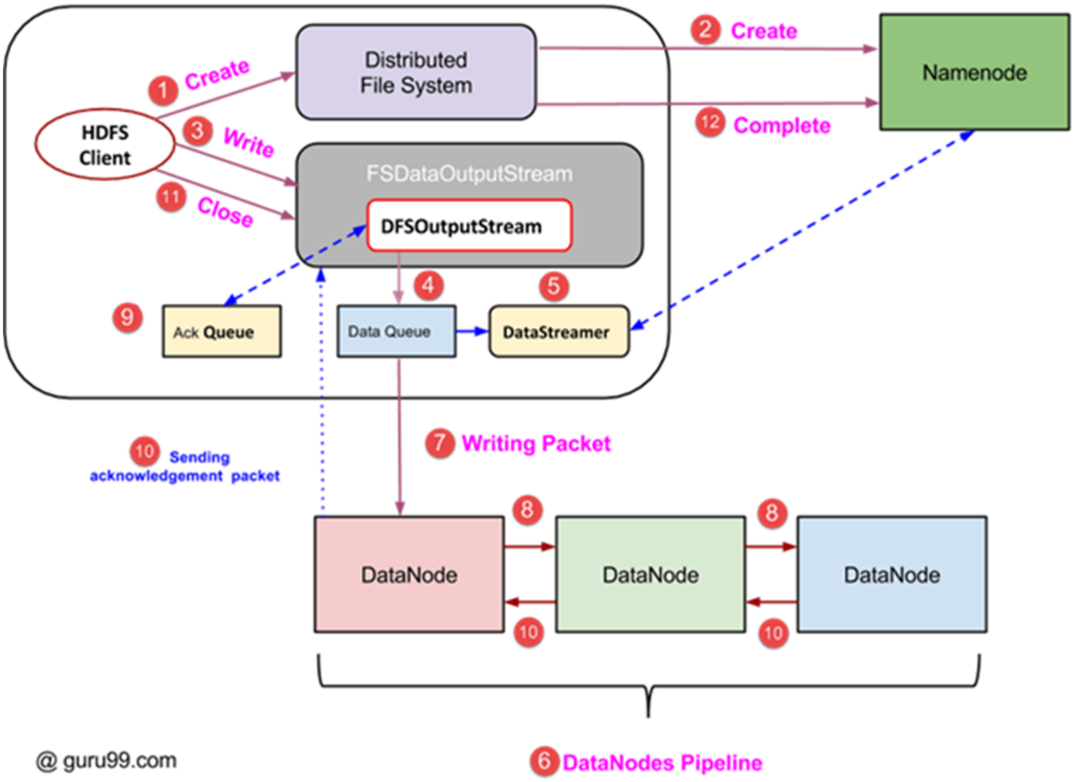


Figure 17

(这俩我真不想看 你要考了我给你跪下了)

※ common command

ls, put, copyFromLocal, get, copyToLocal , cat, tail cd,mkdir?

4.2 distributed database

before

the core purpose of db is to manage the data, relational db has complete relational theory basis , transaction management support and efficient query optimization, index ,synchronization control mechanism

new challenge

concurrency , scalability, availability, question

※ Big Table

use GFS ,as the underlying data storage,

cooperate with MapReduce

use Chubby to provide collaborative service management

can be expanded to PB level

the prototype of HBase

※ HBase

highly reliable , high-performance, column-oriented ,scalable distributed database.

mainly unstructured and semi-structured data.

goal: to process very large tables

1. core component of Hadoop system

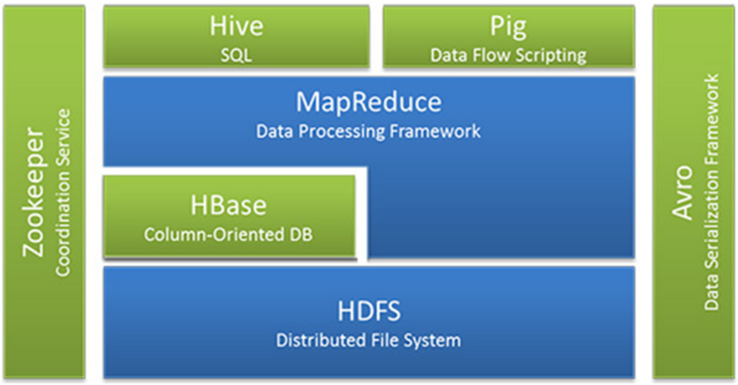


Figure 18

zookeeper to provide collaborative service and error recovery

sqoop provide efficient and convenient relational data import function for HBase

pig and hive: high -level SQL

2. data model

table: consist of multiple rows and colums

row : a row key and one / more columns with values associated with them

column: consist of a column family and a column qualifier

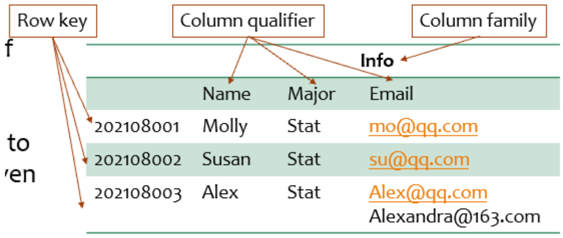


Figure 19

cell: combination of row, column family and column qualifier, contains a value and a timestamp,

timestamp, write along each value, version

HBase is a sparse, multi-dimensional sorted mapping table,

the index is <row key, column family, column qualifier, timestamp>

the value in the table are unexplained string and have no type

The table is composed of one or more column families in the horizontal direction.

A column family can contain any number of columns and the data in same column family is sorted together

why sparse ?

each row of data in the same table can have different columns , for each row of data in the entire table , some columns values are empty, so it is sparse .

Figure 20

the column support dynamic expansion, easily add column without pre-defining

all columns are stored in the type of string update : generate a new version and keep the old version

data storage in order of ts , can choose to get the version closest to a certain time or all version or latest default.

Region: cut

4.3 usage HBase

1. process

set java_home in conf/hbase-env.sh

connect to Hbase

create table , must specify table name and column family name

list information

put data into table

get a single row of data

2. useful command

create list disable enable describe

alter改变一个表 exists drop 从HBase之中删除

4.4 NoSQL database

traditional db show deficiencies in internet application , wave of information.

NoSQL is designed to replace relational db ,

NoSQL is general term for non -relational database .

flexible scalability, data model, close integration with cloud computing

1. typical NoSQL database

key-value , column-oriented, document ,graph database

2. key-value

store the data as key-value sets, where key is unique identifier

value is invisible to the database , can’t be queried and indexed .

key is an object can be any of type

K-V database type : in-memory : stored in memory Redis

Persistent : stored in hard disk berkeley DB

highly partitionable and highly scalable.

3. column-oriented

for distributed storage of massive data ,

use column family database model, the database is composed of multiple rows, each row of data contains multiple column families ,different row can have different number of columns

each row of data is located by the row key

Cassandra, HBase, HyperTable.

4. document database

used to store , retrieve and manage document-oriented information.

document is basic unit of processing information, which is = a record of relational database

aim to store semi-structured data as document , usually use XML JSON and others

locate by keys so is the derivative of KV database

the document database can build an index based on the key or based on the content of document

typical document database include, CouchDB MongoDB RavenDB

document can “self-describe” the type and content of data contained a document can contain very complex data structure, such as nested objects and each document can have a completely different data structure.

5. Graph database

uses Graph as data model , use vertex edge and attribute to represent and store data .

to manage data with a high degree of interrelationship,

social networks , dependency analysis, recommendation system, path finding

Nep4J, infinite Graph, GraphDB

6. 重要！ comparison

	Pros	Cons
Relational database	Based on perfect relational algebra theory, strict standards, support ACID for transactions, efficient querying with the help of indexing mechanism, mature technology, and technical support from professional companies	poor scalability, inability to support mass data storage, inflexible data model, inability to support Web2.0 applications, transaction mechanism affecting the overall performance of the system, etc.
NoSQL database	It can support ultra-large-scale data storage, a flexible data model can well support Web2.0 applications, and it has strong horizontal expansion capabilities, etc.	lack of mathematical theory foundation, poor performance of complex queries, most of which cannot achieve strong transaction consistency, difficult to achieve data integrity, lack of technical support from professional teams, difficult maintenance, etc.

Figure 21

5. ch05 process

Processing architecture

1. centralized processing architecture

data is centrally stored in the central node

all business units of the entire system are centrally deployed on it.

all the function of the system are centrally processed by it.

features: simple, mainframe has excellent performance and good stability , high economic cost, single point problem and poor scalability.

2. distributed processing architecture

use of minicomputer and ordinary PCs to build distributed computing platform at lower cost

feature: economy and maintenance costs are lower, no obvious single point problem, easier to scale horizontally.

Model:

batch processing calculation model : high data throughout rate, suitable for batch processing of massive pre-stored data, typical system such as Hadoop supporting MapReduce model.

stream processing calculation model: processing delay is short , suitable for real-time data stream processing, Strom and S3 platforms

hybrid computing model : integrate the batch and stream but higher system complexity. Spark

Graph processing model: suitable for processing large-scale graph data, Pregel GraphLab.

5.1 batch processing - MapReduce

1. overview

highly abstract complex parallel computing processes running on large-scale clusters into 2 function Map and Reduce.

user don't need to master the detail of the distributed parallel programing , just write Map and Reduce function.

"compute closer to data", find the nearest map node of the data block for calculation, instead of transporting the data to the calculation node

moving data is expensive.

2. divide- and -conquer

use the divide and conquer strategy, divide the large data into many independent chunks and distributed them to multiple nodes for parallel processing.

get the final result by integrating the intermediate result of each node

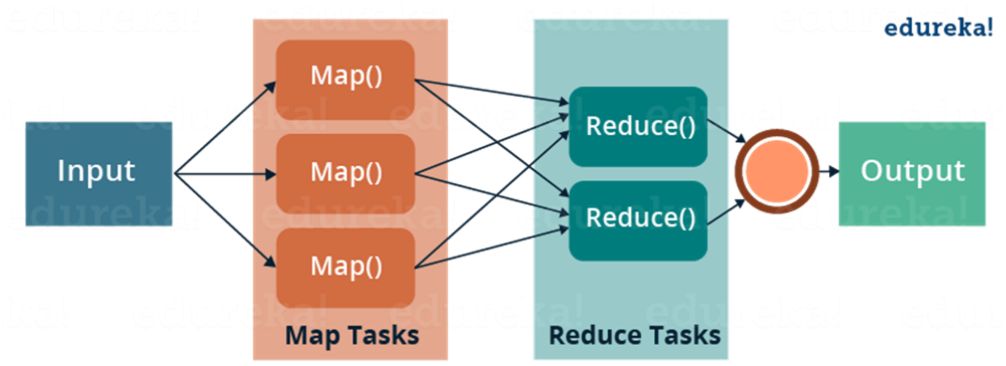


Figure 22

3. example

Map and Reduce take <key,value> as input , convert to another or a batch of <key,value> for output.

4. how it work?

2 nodes : master and slaver

3 execution process : map, shuffle, reduce

Map phase include split and map

split: divide the user data file into multiple chunks, and then copy the user program to the master node and the working node to create the corresponding process

the master node assigns tasks to idle nodes

the worker node assigned the map tasks to read the input data of the chunk. each trunk corresponds to a map task, the worker node executes the map function, calculated a series of intermediate key-value pair results. and write them to memory.

shuffle phase

using the output of the map as the input of reduce

partition, sort and merge the output of the map overflow writing: the result of the buffer will be written to the disk in stages .

when the overflow writing operation is performed, the data will be partitioned. the number of partitions is usually the same as the number of subsequent reduce tasks. each partition is keyed . sort merge the value with the same key at the same time and then write them to disk.

the worker executing the reduce task fetch data from the map side according to the address information provided by the master.

the data may come from 1 or more maps, the data is also cached and then overwritten, merged with the same key value, and then written to disk.

来自同一个map的数据是有序的， reduce -end shuffle

reduce phase

use the ordered files finally generated by shuffle as the reduce input. perform the reduce operation.

the worker executing the reduce task traverses the input file. for the same key, the list of key and associated value is passed to the reduce function, and the output generated will be added to the output file of the partition. when all reduce executed, the master wakes up the user process.

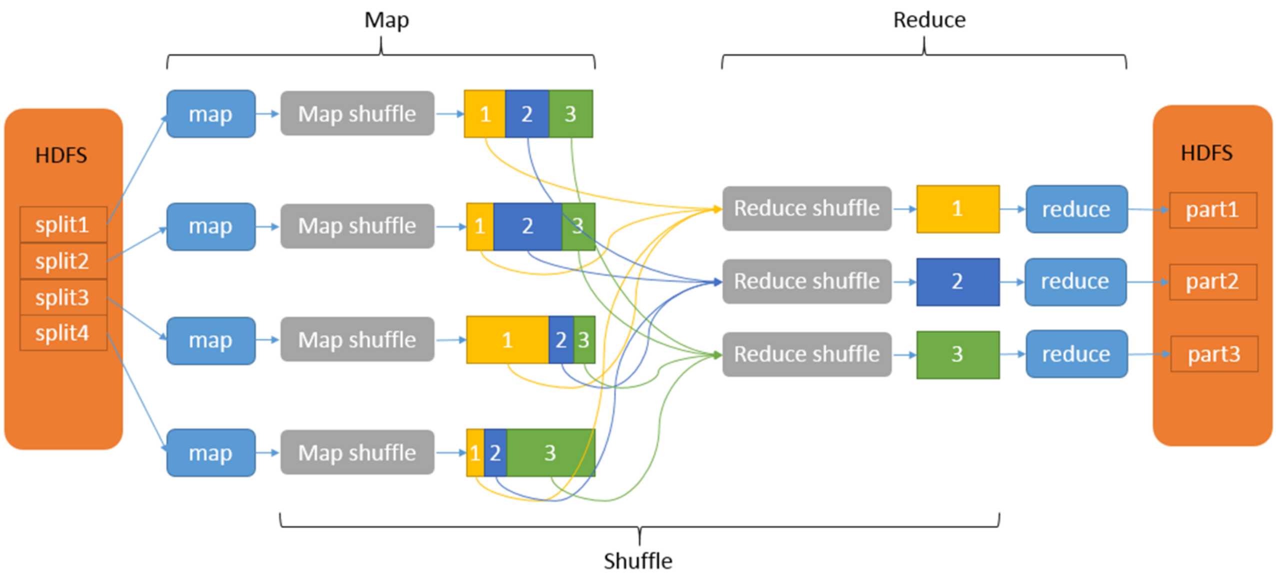


Figure 23

5.2 streaming process- Storm

streaming: real-time acquisition of massive data from different data sources ,through real-time analysis and processing, to obtain valuable information.

Storm:

concepts and topology above

1. stream grouping

determine how Storm routes tuples between tasks in a topology, spout and bolt, different bolts.

when and how a task is sent to tuple is determined by stream grouping

shuffle: randomized round-robin , each task of bolt receive similar number of tuples.

fields grouping: ensure all tuples with same filed value, are always routed to the same task, a simple hash of the field values , modulo to the number of tasks

all grouping: all tuples will route to each task

global grouping: all tuples will route to same task

direct grouping : assign a certain tuple to a certain task

2. components

adopt the master- worker mode

Nimbus is master node, distributes data among all the worker node , assign tasks to worker nodes and monitor faliure

Supervisor is worker node, has multiple worker process, governs worker process to complete the tasks assigned by Nimbus

Worker(process): execute tasks related to a specific topology, create executors and asks them to perform a particular task.

executor(thread) : a single thread spawn by a worker , run one or more tasks but only for a specific spout or bolt.

task: performs actual data processing.

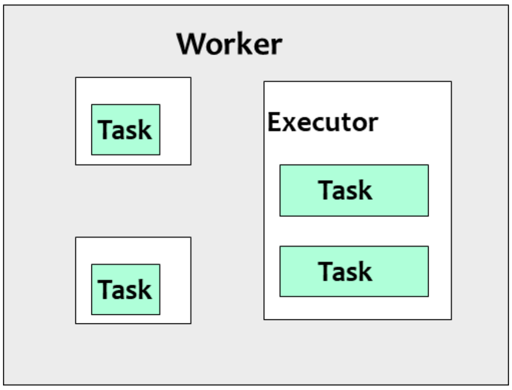


Figure 24

zookeeper is used to coordinate between Nimbus and multiple supervisors.

3. workflow

client submit topology,

Nimbus splits the topology into task chunks and assign them to supervisors. corresponding message are stored in zookeeper.

supervisor receives its own tasks and its worker to process Tasks

worker process creates Executors to run Tasks.

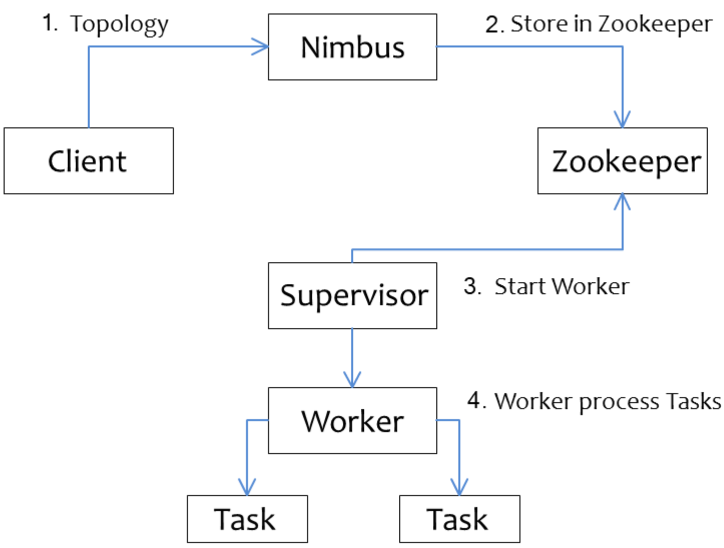


Figure 25

5.3 hybrid processing -Spark

it is a unified analytics engine for large-scale data processing. it provide high-level API in java ,Scala,python and R, and an optimized engine that supports general execution graphs.

breaks the record Hadoop holds on sorting.

1. comparsion

	Hadoop	Spark
Read data	Disk (HDFS)	Memory
Delay	Medium delay	Low delay
Process Unit	Process	Thread
Execution	Iteration	Based on DAG
Scenarios	Batch processing	Hybrid processing, Statistical analysis, machine learning

Figure 26

2. concepts

executor: a process runs on a worker node, responsible for running task

task: a unit of work that sends to the executor

job: the job is parallel computation consisting of multiple tasks that get spawned in response to action in Spark

stage: each job divide into smaller of tasks called stages that depend on each other.

understanding of RDD: contains arbitrary collection of objects. each data set in RDD is logically distributed among cluster nodes so that they can be processed in parallel.

3. workflow

using spark-submit to submit an application

launch the driver program

driver asks for resources to cluster manager

cluster manager launches excutors

driver process run with the help of user application. send work to executors in the form of tasks executor process the task and the result sends back to the driver through the cluster manager.

4. RDD

resilient distributed dataset , the fundamental data structure of Spark.

an immutable (permanent) collection of objects which computes on different node of cluster.

Features:

in memory computation, it stores intermediate results in distributed memory RAM ,instead of stable memory (disk)

lazy evaluation, all transformation in Spark are lazy, they don't compute their results right away. instead, they just remember the transformations applied to some base data set. Spark computes transformations when an action requires a result for the driver program.

fault-tolerance: they track data lineage information to rebuild lost data automatically. each RDD remembers how it was created from other datasets(by transformation like map, join or groupby) to recreate itself.

immutability, data is safe to share across process, can also be created or retrieved anytime which makes caching , sharing and replication easy . consistency

partitioning : fundament unit of parallelism , each one is one logical division of data which is mutable(可变的)

persistence, users can state which RDDs they will reuse and choose store strategy

coarse-grained(粗粒度)operations, it applies to all elements in database through maps or filter or group by operation

location-stickiness, task is close to data as much as possible, DAG

5. RDD operations

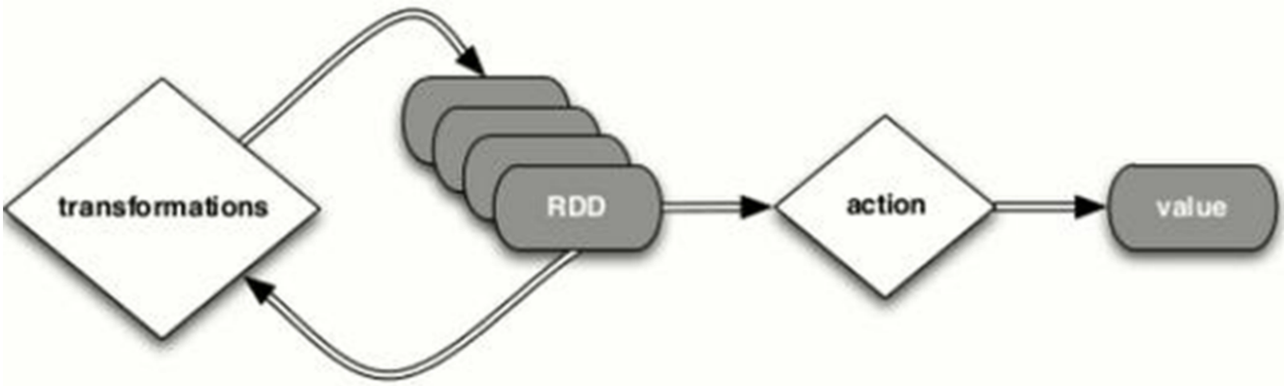
Transformation

functions that take an RDD as the input and produce one or more RDD as the output.

they don't change the input RDD(RDD is immutable, no one can change it), but always produce one or more new RDD by applying the computations they represent. map(), filter(), reduceByKey().

transformation are lazy operation

it create one or many new RDDs, which executes when an action occurs. hence transformation create a new dataset from an existing one.



narrow transformation

the result of map, filter and such that the data is from a single partition only. self-sufficient

spark groups narrow transformations as a stage know as pipelining

wide transformation

the result of groupByKey and reduceByKey like functions

the data required to compute the records in a single partition may live in many partition of the parent RDD

as know as shuffle transformation, they may / not depend on a shuffle .

action

an action in Spark returns final result of RDD computations . brings laziness of RDD into motion

it triggers execution using lineage graph to load the data into original RDD carry out all intermediate transformations and return final results to Drive program or write it out to file system

lineage graph is dependency graph of all parallel RDDs of RDD actions are RDD operations that produce non-RDD value. is a way to send result to the driver. First(), take(), reduce(),collect(),the count() is some of actions in Spark, using transformations, one can create RDD from the existing one, but when we want to work with actual dataset, at that point we use Action.

5.4 usage Spark

code part skip

6. ch06 analysis

6.1 big data analysis

1. intro

big data analysis is the key to transforming big data into useful knowledge,

derived from actual application requirement in practice, driven by specific application data , supported by algorithm ,tools and platforms , used the knowledge and information discovered in practice to achieve a variety of valuable service (md 看半天全没用)

why big data analytics important?

cost reduction, faster and better decision making and new products and services .

2. algorithm

simple analysis

use conventional(传统的常规的) method on OLAP (这词你也得查一下也是无敌了)

use SQL to achieve statistical analysis results .

but simple can not get the in-depth value.

ML and DL algorithm

Types	Problems	Typical algorithms
Supervised learning	Classification Regression	Naïve Bayes, Decision Tree Linear regression
Unsupervised learning	Clustering Association rule learning Anomaly detection	K-means, DBscan, EM Apriroi algorithm K-NN, LOF
Semi-supervised learning	Generative models	
Deep learning	Convolutional neural networks Autoencoder Generative adversarial networks Recurrent neural networks	CNN Variational Autoencoder GAN RNN, GRU, LSTM
Ensemble learning	AdaBoost, Gradient boosting machine	XGB, CatBoost
Dimension reduction	Principal component analysis	PCA, PCR

Figure 27

6.2 big data learning

1. distributed machine learning DML

an algorithm or system that uses multiple working nodes for machine learning or deep learning.

use computer clusters to achieve large-scale machine learning model training.

adopt multi-threaded or multi-machine parallel computing based on shared/virtual memory

divides data and distributes to multiple working node

each working node trans the models according to the local data

do integration to get global model

2. DML module

Data and Model split module, with no shared memory or data is too large and cannot be stored in the local memory, data is divided and distributed to each working node.

Single Machine Optimization Module, each node is trained based on the local data, and sub-models assigned to it

Communication Module, content of communications is mainly the sub-model parameters or parameter update information

Aggregation Module, a global model is generated by aggregating local data/ sub-models from different working nodes

3. Federated Learning (联邦学习)666 限时返厂

also known as collaborative learning.

machine learning technique that trains an algorithm across multiple decentralized edge devices or servers holding local data samples, without exchanging them.

enable multiple actors to build a common , robust machine learning model without sharing data.

address critical issues such as data privacy ,data security , data access rights and access to heterogeneous data (异质)

4. data representation

represent the data Di as a matrix , each row indicates ad a data sample , each column indicated a feature. 横向表示用户维度，纵向表示特征维度

feature space X, label feature space Y, data sample ID space I, so the training data set is (I,X,Y)

此处的分类都是说两个数据集之间的重叠如何，

categories :

horizontal: 用户特征空间重叠较大

also knows as federated learning by samples.

data sets of participants have overlapping data features , but data samples owned are different.

the branches of a banking system. two bank’s business is similar the feature is similar but the id is not.

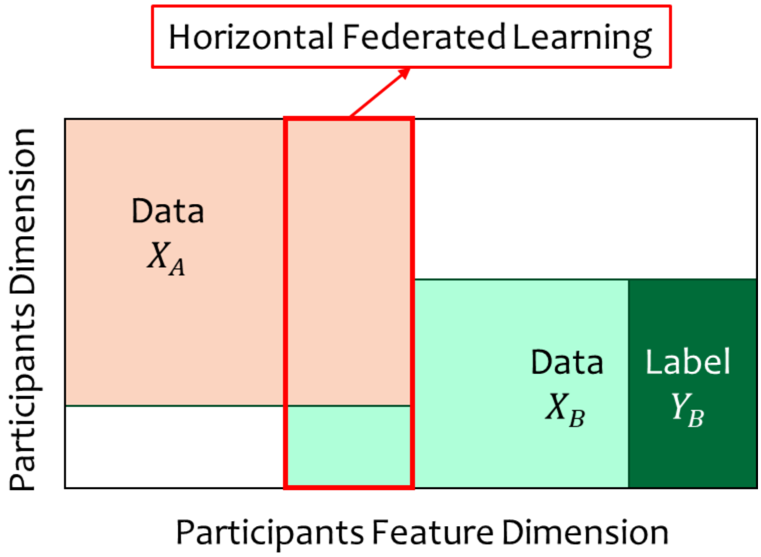


Figure 28

vertical

also known as federated learning by feature

the participants have overlapping data samples , but the data features are quite different.

two institution in the same area, one has user's consumption records, the other has user's bank record, have many overlapping sample, but the data characteristics are different.

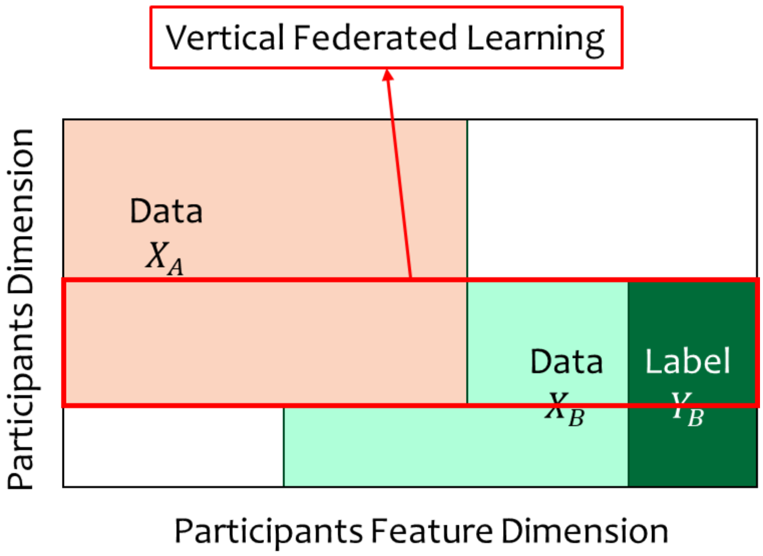


Figure 29

transfer

the data features and samples of two participants have little overlap.

two organizations in different regions, one own consumption record and other has the bank records , nothing in overlap

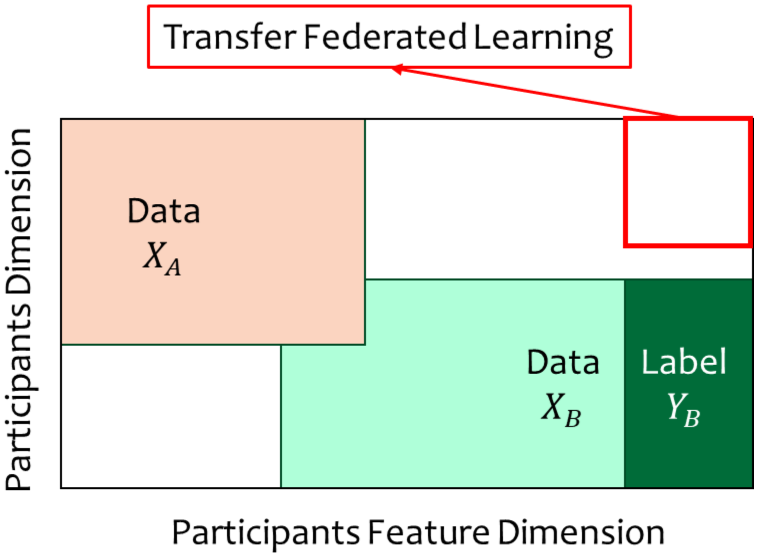


Figure 30

5. client - server architecture

typical horizontal federated learning architecture.

k participants with the same data structure train a ML model together with the help of a server. (also, called a parameter server or an aggregation server)

training step(大数据技术导论考了这个题xs)

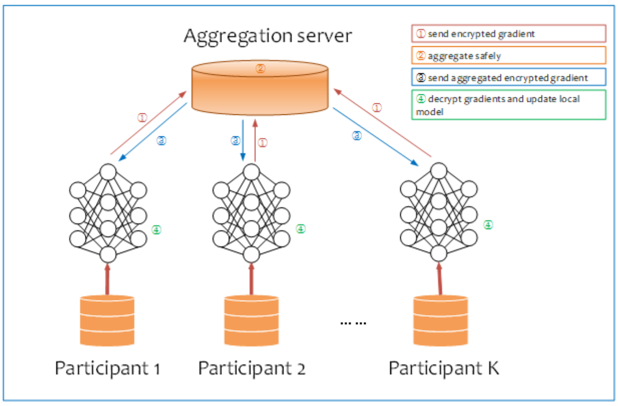


Figure 31

Step 1: Each participant calculates the model gradient locally and uses homomorphic encryption(同态加密), differential privacy or encryption sharing technology to encrypt, and sends the model gradient to the server after encryption;

Step 2: The server performs secure aggregation without knowing any participant information;

Step 3: The server sends the aggregated result back to each participant;

Step 4: Each participant decrypts the received aggregate gradient and uses the decrypted gradient result to update their model parameters.

6. Spark MLlib

machine learning library, a scalable machine learning library that discuss both high-quality algorithm and high speed.

DataFrame-based API is the primary Machine Learning API

6.3 big data visualization

an interdisciplinary(还记得这个词什么意思吗) field that deals with graphic representation of data.

a map between the original data and graphic elements.

into vivid and easy-understand way graphics, images, maps, animations.

1. effect of visualization

tracking data, finding hidden pattern, promote comprehension, analysis and interface.

2. categories

Graphic visualization

node link graph, not suitable for dense graph

simplification: cluster visualization, edge binding, adjacent matrix, GMap.

Multi-dimensional data

data variables with multiple dimensional attributes.

parallel coordinate system, scatter plot matrix, dimensionality reduction,

Spatial-temporal Data

geographic and timestamp

Text visualization

word cloud, semantic structure visualization 语义结构 ,text dynamic visualization

Interactive visualization

assist users in analyzing and reasoning large-scale complex data sets through an interactive visual interface.

representation and interaction are 2 main components of data visualization.

3. Tools

燃尽了

7. ch07 security

7.1 security challenge

1. double-edged sword 双刃剑

brings privacy and security challenge.

Figure 32

2. privacy

narrow sense: personal contact information , friend relationship information, private information(age, salary,occupation),

broad sense: information scattered (散布) on all corner of internet, search engine search history, browser traces, shopping record of e-commerce platform, search history of the map application.

3. concept

Data anonymization (数据匿名化): to hide some sensitive information in the database, making it difficult for the data subject to be identified.

the owner of personal information is called the data subject(数据主体), the data controller attempt to protect the privacy of the data subject through anonymized data.

reason for anonymization, storage value will be greatly reduced.

opposite of anonymization, deanonymization,

示例

AOL-> remove id is not anonymization

long-tail 非常规喜好可能成为去匿名化的线索

7.2 data privacy techniques

1. relational data example

post SSN->Salary will infringe personal privacy.

remove social security number to create de-identified data, whether the de-identified form protect personal privacy depends on information known to attacker. 去标识化并不意味着安全

cannot uniquely identify anyone’s salary, date of birth is the Quasi-identifier (准标识符)

quasi-identifier : date of birth, gender.

2. link attack

quasi-identifier 组合起来可以唯一定位个体。

3. example of anonymization

tuple suppression: replace some value to * 增加uncertainty, but loss of information.

generalization: 把具体值替换为更粗的类别，保留一定的可用性

SA replacement/ disturbance: 保持记录可识别但让敏感之变得模糊或者加噪声（替换为区间）

4. K-Anonymization

一个被发布的数据集满足K-Anonymization, 意味对于任意一个记录，其在准标识符上的值至少与其他k-1条记录相同，也就是每个人在至少k人里不可被区分

防止QI的link attack

实现方法：

suppression: 把部分值替换为 *

generalization：把具体值替换为更宽泛的类别

5. uncertainty in K-Anonymization

k-anonymization 给出的是一种不确定性集合，发布的匿名表T' 实际上对应着许多可能的原始表。

对查询的响应会受到这种不确定性的影响，例如：查询到某具体的工资是多少，因为这个组合对应k个记录,最佳猜测可能是这些记录的加权平均，而对于最大最小问题，答案会被扩展为一个区间

（之后再看一遍PPT）

8. ch08 application

以业务逻辑出发，business logic 还是以数据为中心

8.1 recommendation system

1. biomedical science

epidemic prediction,

smart medical, promote sharing of high-quality medical resource and avoid repeated patient examination.

bioinformatics : in-depth understanding of biological process and disease-causing genes

2. logistics

intelligent logistics based on big data and internet of things technology improve the level of logistics informatization and intelligence

reduce logistics cost

improve logistics efficiency

8.2 smart city

1. intelligent transportation

real-time traffic monitoring, traffic intelligent guidance , public vehicle management

2. environment monitoring

monitor and analyze air/water pollution

3. urban planning
4. security field
5. economics
6. vehicle
7. telecom industry 电信
8. energy industry
9. sports

8.3 more

1. recommendation system

Search engine can only solve clear needs , the recommendation system is a typical application of big understand the user's preferences by analyzing the user's historical records , actively recommend the user's interested information to meet the user's personal recommendation need, a tool that automatically contacts user and item

Long Tail

argue that products in low demand or that have a low sales volume can collectively build a better market than their rivals retailing strategy of selling a large number of different items which each sell in relatively small quantities, usually in addition to selling large quantities of small number of popular item

you can increase sales by discovering long-tail products and recommending them to the interested users.

How to recommend

establish the connection between the user and the item

1. expert : senior professionals will recommend item manually
2. statistics-based: based on statistical information
3. content based : use machine learning methods to describe the characteristics of the content, and find similar content based on it
4. collaborative filtering, most successful, using product evaluation information of users similar to the target user to predict the target user's preference for a specific product
5. hybrid : combine multiple

System model

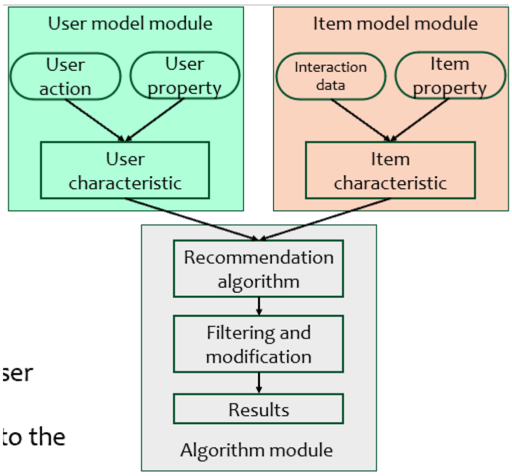


Figure 33

user: analyze user interests and needs based on user behavior data and user attribute data

item: based on object data

algorithm module : calculate the objects that the user may be interested in , results are adjusted according to the recommendation scenario

8.4 GFT google flu trend

provide estimates of influenza activity

aggregating google search queries , attempt to make accurate predications

take advantage of big data analytics

correlations : search key words + influenza incidence

use color + time dimension 颜色加时间维